SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems

COURSE CODE: CSA0509

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COURSE OUTCOME COVERED

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CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool: Concept Mapping (Medium)

Assessment Tool Weightage: 20%

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Code of Conduct:

I, Judy Allen J (192520055) (Name / Reg No) certify that this submission is my

original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student:

Judy Allen J

Q1. Construct a concept map linking the following terms:

DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship

```
                    +----------------+
                    |    DBMS     |
                    +----------------+
                      /      |       \
                     /       |uses   \manages
                    /        |        \
                   /         |         \
          +------------+   +-----------+   +-----------+
          |  Schema  |   | Data Model|  | Instance  |
          +------------+   +-----------+   +-----------+
                |                 |
            describes     represented by
                |                 |
          +------------+          |
          | Database  |           |
          |  Structure |          |
          +------------+          |
                                  |
                                  |
                    +---------------+
                    |  ER Diagram   |
                    +---------------+
                       /       \
                 contains      contains
                     /           \
             +-------------+    +--------------+
             |  Entity   |<>----<Relationship >
             +-------------+    +--------------+
                   |
                 <has>
                   |
              ( Attribute )
```

Q2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.

```
                    +----------------------+
                    |    DATABASE         |
                    +----------------------+
                               |
        ------------------------------------------------
        |                      |                      |
        |                      |                      |
+---------------+      +---------------+      +----------------+
| External Level|  -> | Conceptual   |  ->  | Internal Level |
| (User Views)  |     |     Level     |      | Physical Data  |
+---------------+      +---------------+      +----------------+
        |                      |                      |
   User Interface       Logical Structure      Physical Storage
```

External ↔ Conceptual = Logical Data Independence

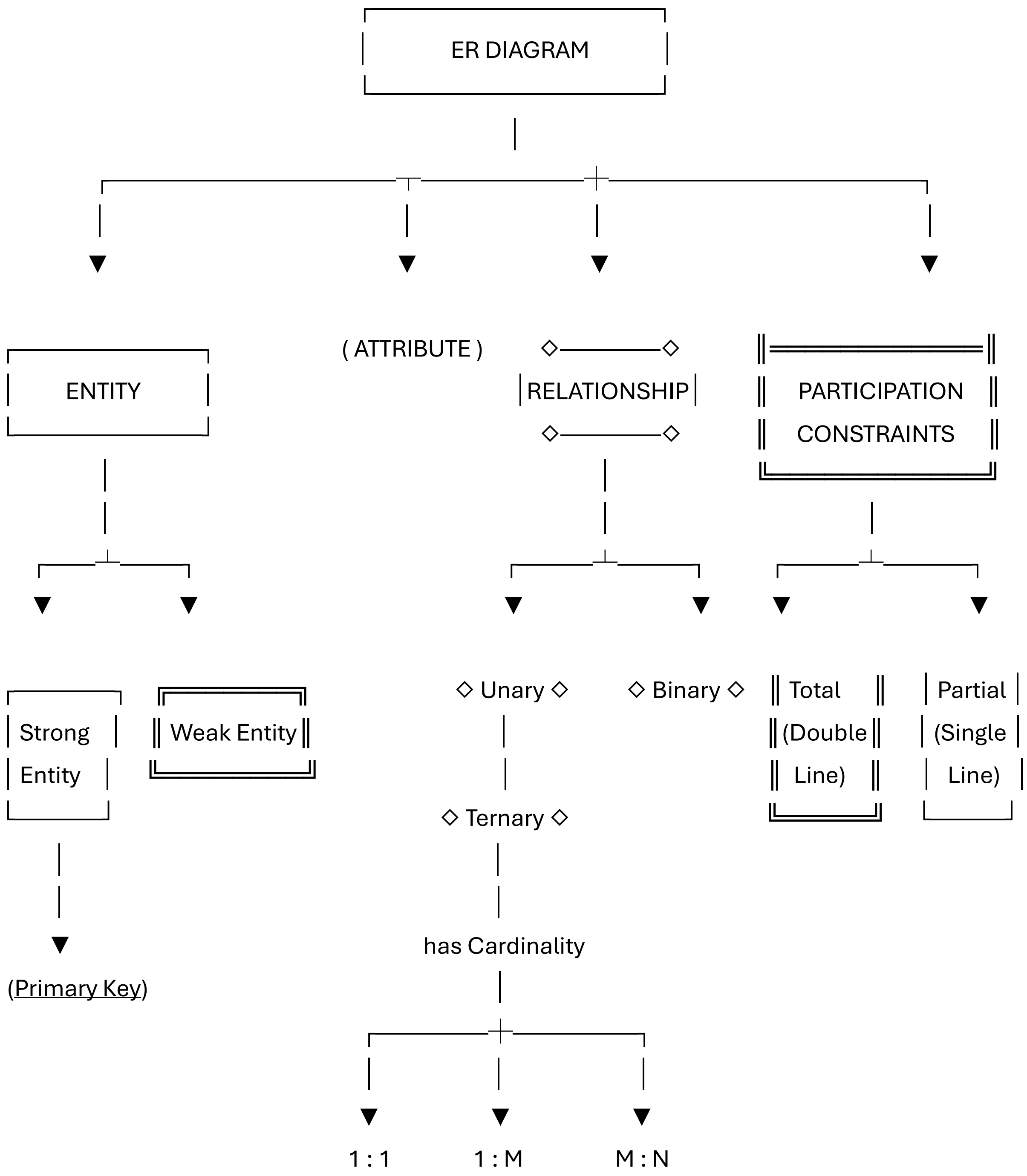
Conceptual ↔ Internal = Physical Data Independence

Q3. Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.

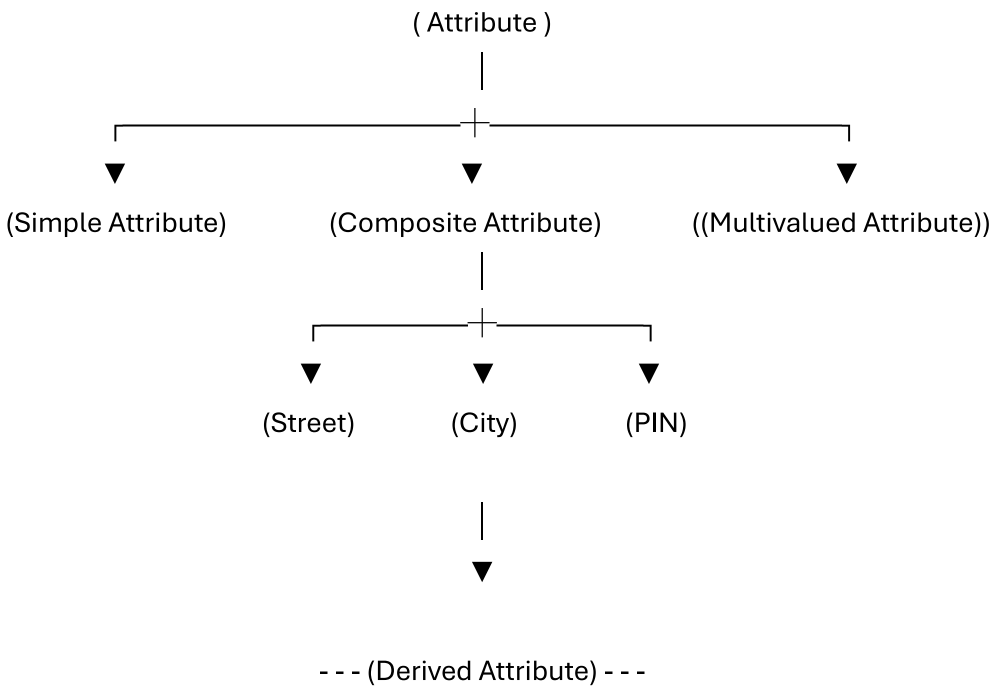
```
      +----------------+
    | DATA STORAGE   |
      +----------------+
            /      \

           /        \
```

File System		DBMS
Redundancy		Reduced Redundancy
Inconsistency		Consistency
No Security		Security
No Backup		Backup
Poor Sharing		Data Sharing

Q4. Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.



ATTRIBUTE TYPES



Q5. Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.

DATA MODELS

consists of

HIERARCHICAL MODEL

Structure

Tree Structure (Parent-Child)

Relationship

One Parent Only
One Child Path

Advantages

- Simple Design
- Fast Access

NETWORK MODEL

Structure

Graph Structure (Many-to-Many)

Relationship

Multiple Parents
Multiple Paths

Advantages

- Flexible Links
- Reduced Redundancy
- Efficient Navigation

RELATIONAL MODEL

Structure

Tables (Rows & Columns)

Relationship

Primary Key &
Foreign Key

Advantages

- Easy SQL Queries
- Data Independence
- Easy Maintenance

Disadvantages

- Rigid Structure
- No Many-to-Many

Applications

File Systems
Organisation Charts

Disadvantages

- Complex Design
- Difficult to Manage

Applications

Airline Reservation
Telecom Networks
Social Networks

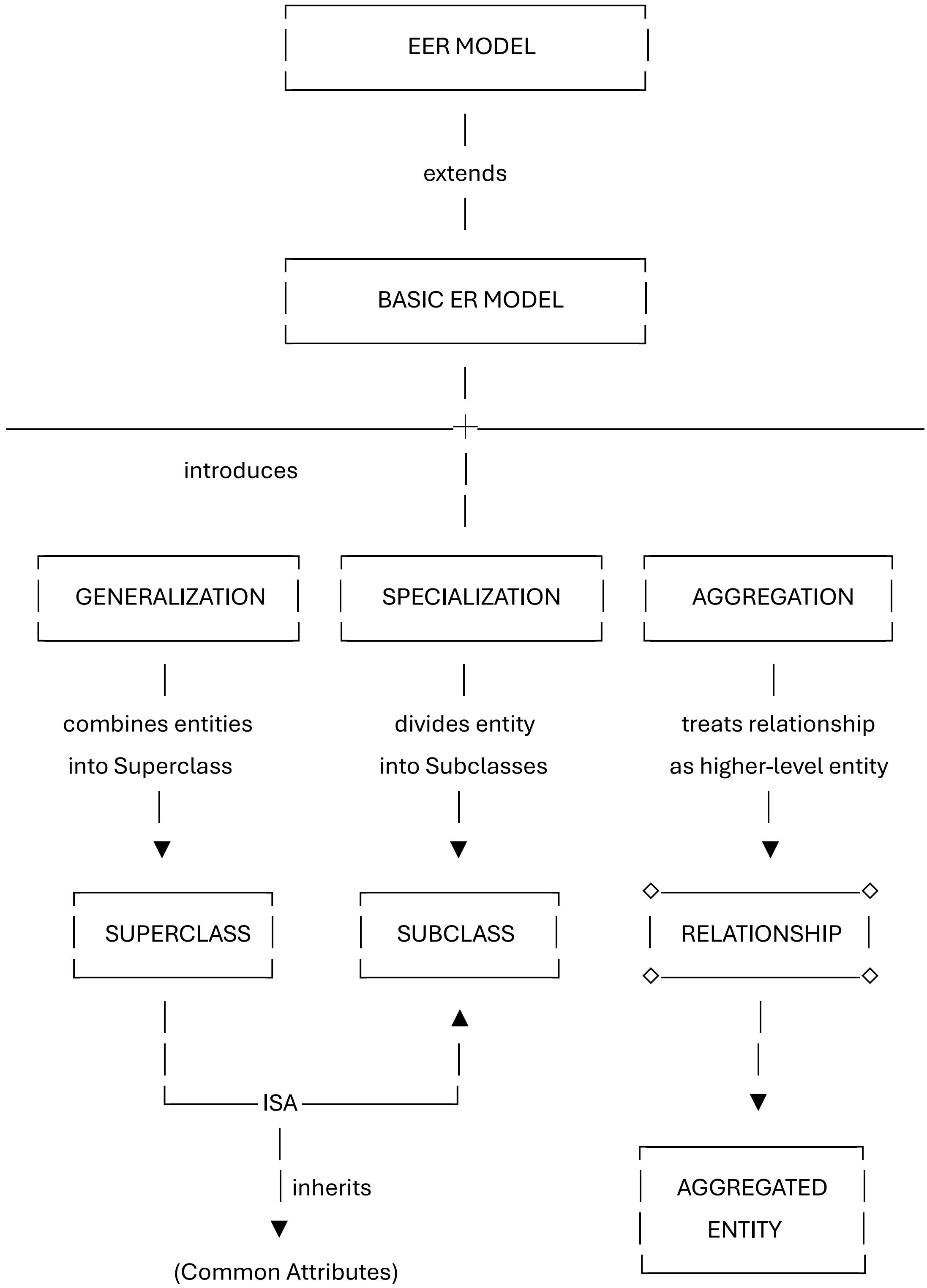
Disadvantages

- Join Operations
- Slightly Slower

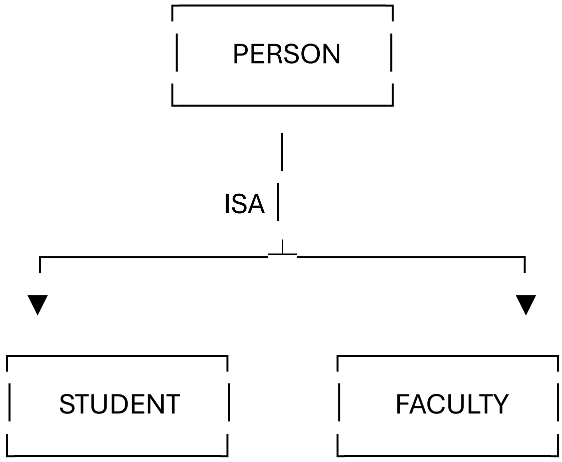
Applications

Banking
Hospitals
Colleges

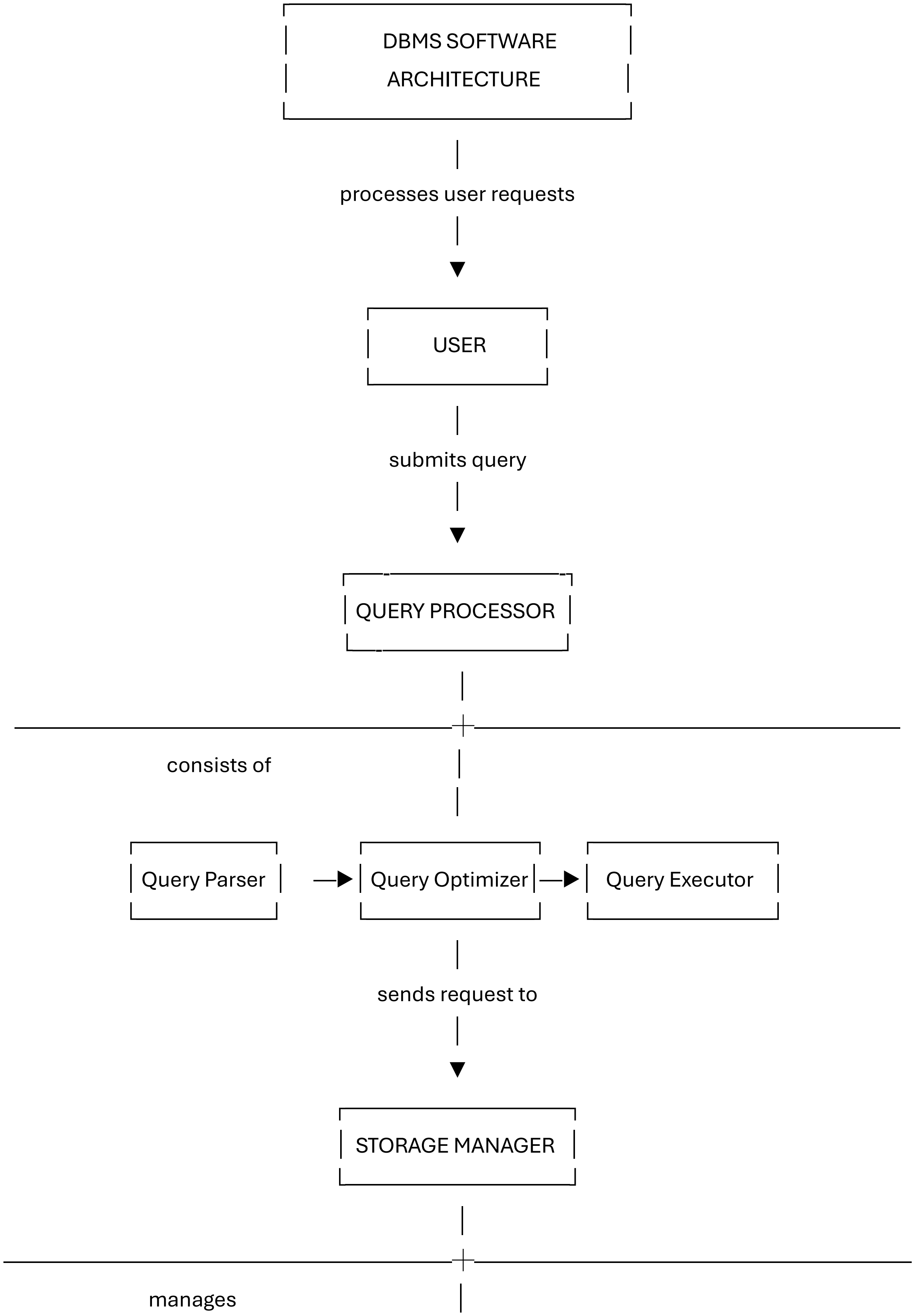
Q6. Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.



Example



Q7. Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.



```
                                   │
┌──────────────────┐ ┌──────────────────┐ ┌──────────────────┐ ┌──────────────────┐
│  Buffer Manager  │ │   File Manager   │ │ Transaction Mgr. │ │ Authorization Mgr│
└──────────────────┘ └──────────────────┘ └──────────────────┘ └──────────────────┘
                                   │
                             accesses data
                                   │
                                   ▼
                     ┌──────────────────────────┐
                     │      DATABASE FILES      │
                     │ Tables • Indexes • Logs  │
                     └──────────────────────────┘
                                   │
                           retrieves records
                                   │
                                   ▼
                     ┌──────────────────────────┐
                     │   QUERY RESULT / DATA    │
                     └──────────────────────────┘
                                   │
                           returns result to
                                   │
                                   ▼
                           ┌──────────────┐
                           │     USER     │
                           └──────────────┘
```

Q8. Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.

DATA MODELS

consists of

HIERARCHICAL MODEL

organized as

Tree Structure

(Parent → Child)

suitable for

File Systems

Organisation Charts

XML Documents

because it

- One Parent
- Fast Data Access
- Simple Structure

NETWORK MODEL

organized as

Graph Structure

(Many-to-Many)

suitable for

Airline Reservation

Social Networks

Telecommunication

because it

- Multiple Relationships
- Flexible Navigation
- Efficient Connections
- Complex Data Handling

RELATIONAL MODEL

organized as

Tables (Rows & Columns)

(Primary & Foreign Keys)

suitable for

Banking Systems

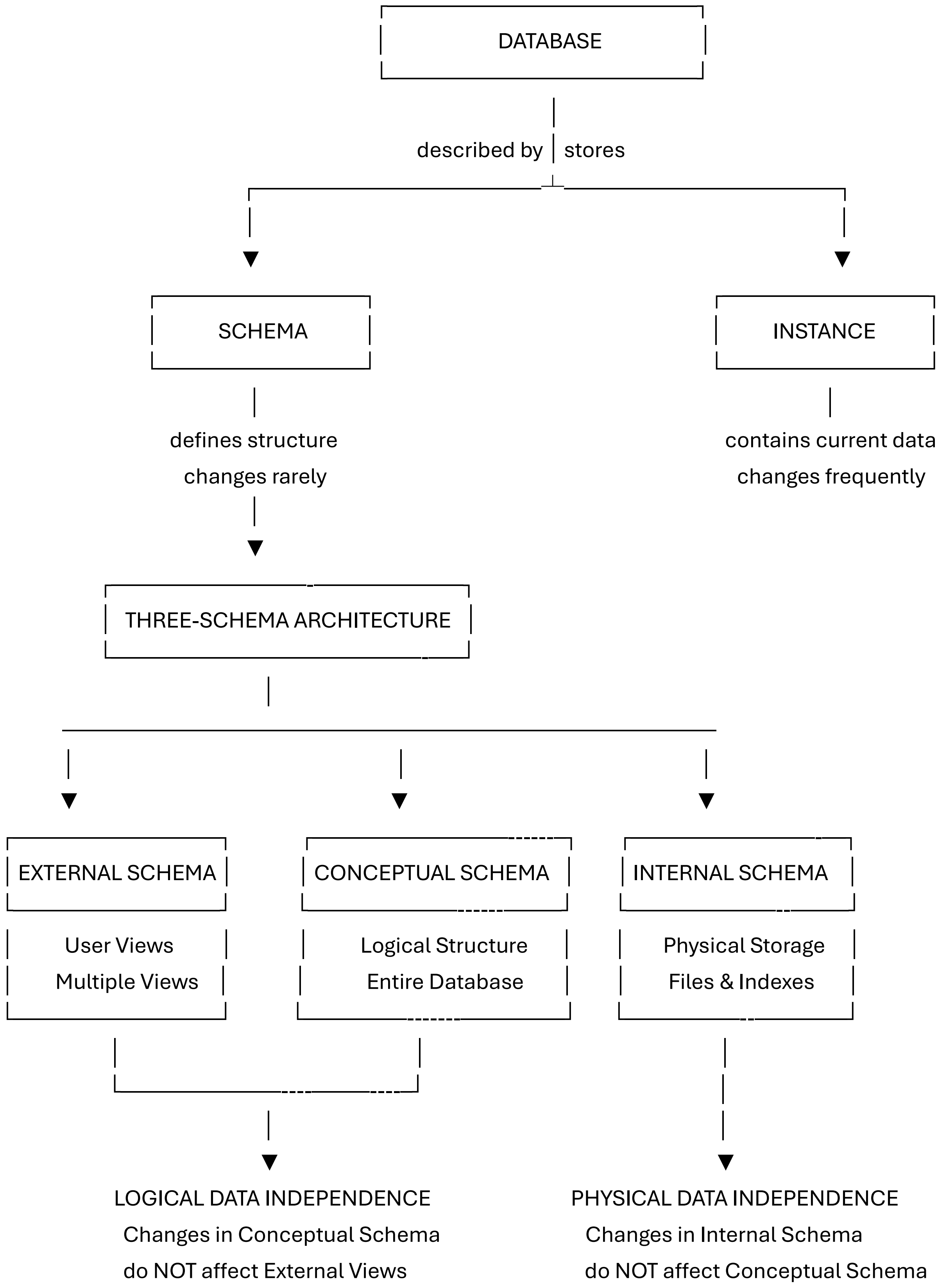
Hospital Systems

College Databases

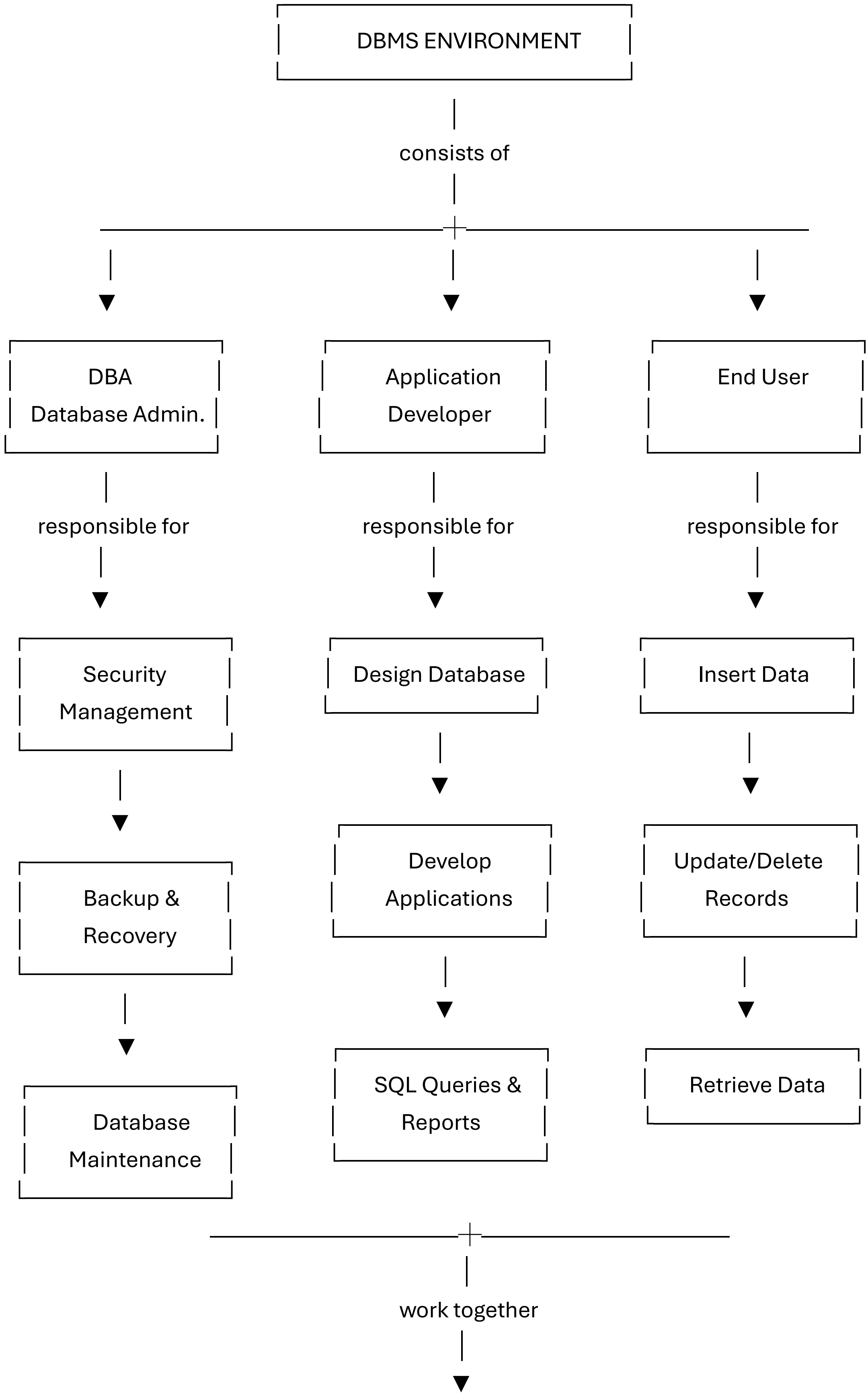
because it

- Easy SQL Queries
- Data Integrity
- Reduced Redundancy
- Easy Maintenance

Q9. Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.



Q10. Design a concept map for the roles and responsibilities in a DBMS Environment: DBA, End User, Application Developer.



Efficient Database Management

ensures

▼

Data Security • Data Integrity • Data Availability

Efficient Processing • Reliable Database Performance